

1.6. Residue class rings

Rings to keep in mind:

$$\mathbb{Z} \quad \mathbb{K}[x]$$
$$\mathbb{K}[x_1, \dots, x_n]$$

Def 1.6.1 Let R and S be rings. A map $\varphi: R \rightarrow S$ is called a ring homomorphism if: $\varphi(a+b) = \varphi(a) + \varphi(b)$, $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$, $\varphi(0) = 0$, $\varphi(1) = 1$ for all $a, b \in R$.

If φ is bijective, then φ is called a ring isomorphism and the respective rings R and S are called isomorphic rings. 1
additionally notation $R \cong S$. Informally, isomorphic mean "the same up to 'translation'". The kernel of φ is $\ker \varphi = \{a \in R : \varphi(a) = 0\}$. this one is redundant. Why?

Rem 1.6.2. injective homomorphisms are called monomorphisms; surjective homomorphisms are called epimorphisms, but we avoid using these extra words.

Examples 1.6.3.

(1) $\varphi: \mathbb{R}[x, y] \rightarrow \mathbb{R}[x]$, $\varphi(f) = f(x, x^2)$. $\varphi(x^2 - y) = x^2 - x^2 = 0$ check in fact, with more work/theory, one can show that $\ker \varphi = \langle x^2 - y \rangle$.

(2) $\varphi: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$, $\varphi(f) = f(x+1)$. This is clearly a ring homomorphism. Is it an isomorphism?

It is: $\varphi^{-1}(f) = f(x+1)$ easy to check that this in fact correct $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \text{id}$
↑
identity map.

(3) $\varphi: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$, $\varphi(f) = f(x^2)$. Clearly, φ is a ring homomorphism.

Surjective? Not surjective, because $g(x) = f(x)$ is an even polynomial: $g(x) = g(-x)$.

Injective? It is injective, because $\varphi(f)$ "knows" the value of f on $\mathbb{R}_{>0}$ and this is enough to know f .

(2)

Rem 1.6.4. The kernel of a ring homomorphism $\varphi: R \rightarrow S$ is always an ideal:
 $a \in \ker(\varphi), b \in R \Rightarrow \varphi(a) = 0, b \in R \Rightarrow \varphi(a \cdot b) = \varphi(a) \cdot \varphi(b) = 0 \cdot \varphi(b) = 0$

$\Rightarrow ab \in \ker(\varphi)$. In fact, more is true: every ideal is a kernel

of some ring homomorphism.

Observation 1.6.5. A ring homomorphism $\varphi: R \rightarrow S$ is injective $\Leftrightarrow \ker \varphi = \{0\}$.

Proof: Necessity is clear. Let's verify the sufficiency. Assume $\ker(\varphi) = \{0\}$. Then,
 $\varphi(a) = \varphi(b) \Rightarrow \varphi(a - b) = 0 \Rightarrow a - b \in \ker \varphi = \{0\} \Rightarrow a = b$. $\square$

Def 1.6.6. For an ideal I in a ring R elements $a, b \in R$ are called congruent modulo I (notation: $a \equiv b \pmod{I}$) if $a - b \in I$. More specifically, for a principal ideal $I = fR$ ($f \in R$), one uses $a \equiv b \pmod{f}$ to denote $a \equiv b \pmod{fR}$, this is called congruence of a and b modulo f .

Def 1.6.7. For a ring R , the ideal $I = R$ is called the unit ideal; it is the only ideal that contains 1 as an element.

Def 1.6.8 (Residue class rings) For a ring R , $I \subset R$ an ideal I , the set $[u]_I := u + I = \{u + r : r \in I\}$ is called the residue class (or coset) of u modulo I . Clearly, $[u]_I = \{v \in R : u \equiv v \pmod{I}\}$. We denote by R/I the set of all cosets of elements of R modulo I and introduce the addition and multiplication on R/I by $[a]_I + [b]_I := [a + b]_I$ and $[a]_I \cdot [b]_I := [a \cdot b]_I$. These operations are well defined. That means: $a \equiv a' \pmod{I}$, $b \equiv b' \pmod{I} \Rightarrow a + b \equiv a' + b' \pmod{I}$, $a \cdot b \equiv a' \cdot b' \pmod{I}$. The structure $(R/I, +, \cdot)$ is called the residue class ring or the factor ring or the quotient ring of R modulo I . The zero of R/I is $0_{R/I} = [0]_I$ and the one of this structure is $1_{R/I} = [1]_I$.

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Remark: R/I is a unitary commutative ring (or what is called the zero ring in other sources) & if $I = R$, then R/I consists of just one element, which is both 0 and 1).

If I is a principal ideal, $I = fR$ for some $f \in R$,

then we also use the notation $R/f := R/fR$ and $[u]_f := [u]_{fR}$.

Examples 1.6.9

(1) $\mathbb{Z}/2\mathbb{Z} = \mathbb{Z}/2$ is the field $\mathbb{F}_2 = \{0, 1\}$, where $1+1=0$.

$$0_{\mathbb{F}_2} = [0]_2, \quad 1_{\mathbb{F}_2} = [1]_2$$

(2) $\mathbb{Z}/3\mathbb{Z} = \{[0]_3, [1]_3, [2]_3\}$ is the field $\mathbb{F}_3$

(3) $\mathbb{Z}/6\mathbb{Z} = \{[0]_6, [5]_6\}$ not a field.

(4) $\mathbb{R}[x]/(x^2+1)\mathbb{R}[x] \cong \mathbb{R}[\sqrt{-1}] = \mathbb{C}$, the field of complex numbers

④

$$(5) \mathbb{R}[x, y] / \langle y - 2x \rangle \cong \mathbb{R}[x]$$

(5)

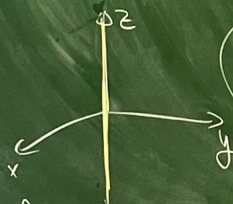
(6) $\mathbb{R}[x] / \langle x^2 - 1 \rangle \cong \mathbb{R}[\alpha]$ where $\alpha = [x]_{x^2-1}$, that is, α is a formal element that satisfies $\alpha^2 - 1 = 0$ that has been added to $\mathbb{R}$ from the "outside". For example you consider some polynomial, say $p(x) = 3x^3 - x + 1$. Take $[p]_{x^2-1} = [3x^3 - x + 1]_{x^2-1} = [3x^3]_{x^2-1} - [x]_{x^2-1} + [1]_{x^2-1} = 3[x]_{x^2-1}^3 - [x]_{x^2-1} + 1 = 3\alpha^3 - \alpha + 1$, just the same any polynomial, so $\mathbb{R}[x] / \langle x^2 - 1 \rangle = \mathbb{R}[\alpha]$. $\leftarrow$ is it a field? $(\alpha-1)(\alpha+1) = \alpha^2 - 1 = 0$
 $\Rightarrow \mathbb{R}[x] / \langle x^2 - 1 \rangle$ is not a field. If one uses the Chinese Remainder Theorem (to come), one will be have in sight into the structure of $\mathbb{R}[x] / \langle x^2 - 1 \rangle$; as a byproduct, one will see that it is not a field.

$$(7) \mathbb{R}[x, y, z] / \langle x, y \rangle \cong \mathbb{R}[x, y, z] / I(V(x, y))$$

$$(8) \text{ For } X = \mathbb{A}^n, \quad \cong \mathbb{R}[z]$$

then $k[x_1, \dots, x_n] / I(X)$ the coordinate ring of X ;

$$(9) \mathbb{R}[x, y] / \langle x^2 + y^2 - 1 \rangle = \mathbb{R}[\cos t, \sin t]$$



$$(10) \mathbb{Q}[x] / \langle x^2 - 2 \rangle \cong \mathbb{Q}[\sqrt{2}]$$

It is a field, Try to see why.

$f \equiv g \pmod{I(X)}$ means f and g get evaluated to the same function on X .

Def 1.6.10. If I is an ideal in a ring R , then the map (6)
 $\varphi: R \rightarrow R/I$, $\varphi(x) = [x]_I$, is a ring homomorphism
called the canonical homomorphism for the residue class ring R/I .
Its kernel is exactly I .

Def 1.6.11. For ideals I, J in a ring R , with $I \subseteq J$, the map
 $\varphi: R/I \rightarrow R/J$, $\varphi([u]_I) = [u]_J$ is well defined.
It's a ring homomorphism and it's called the canonical homomorphism
from $R/I \rightarrow R/J$.

Example 1.11.

$$(1) \mathbb{Z}/6 \rightarrow \mathbb{Z}/3, [z]_6 \mapsto [z]_3 \quad (6\mathbb{Z} \subseteq 3\mathbb{Z})$$

$$(2) \mathbb{R}[x]/\langle x^2-1 \rangle \rightarrow \mathbb{R}[x]/\langle x-1 \rangle, [f]_{x^2-1} \mapsto [f]_{x-1}$$